Climate Change, Science and Policy

40 key historical facts stated in Jeff Kisling's blog posts on this topic, each checked against an outside source.

Climate Change, Science and Policy

Jeff's posts on climate change, mostly from 2017–2022, follow the science (CO2 records, IPCC and U.S. assessments, temperature records) and the politics (the Paris Agreement, the Green New Deal and the Sunrise Movement, the Inflation Reduction Act, and continued fossil fuel expansion and financing). They also track extreme weather close to home, such as the 2019 Missouri and Mississippi River floods and the 2020 Iowa derecho, and disasters worldwide.

40 facts checked against outside sources, 3 of them corrected (the posts had a detail slightly wrong; the corrected version is shown). Each fact lists the source that confirms it and the posts that state it. The wording is a plain summary, not a quotation from the posts.

Facts

1. 2008-10-10 (corrected)

In an October 2008 address, a UNHCR senior adviser on climate change said current estimates of people who would be forcibly displaced by climate change by 2050 ranged from 250 million to one billion.

> ***Correction:*** *Posts say UNHCR estimated up to 250 million; the UNHCR document gives 250 million as the low end of a range of estimates (250 million to one billion).*

- **Source:** Address by Mr. Yoichiro Tsuchida, UNHCR Senior Advisor on Climate Change, Bonn, 10 October 2008
- **In the posts:** Design and Build Beloved Community Models (2018-02-22); What is our response to environmental chaos? (2018-05-29); Prefab Communities (2021-07-01)

2. 2015-11

In November 2015 New York Attorney General Eric Schneiderman subpoenaed Exxon Mobil, investigating whether the company misled the public and investors about the risks of climate change.

- **Source:** NY attorney general investigates Exxon Mobil over climate claims (Fortune)
- **In the posts:** An Evolving View of Climate Change (2015-11-12)

3. 2015-12-12

The Paris Agreement was adopted by 195 Parties at COP21 in Paris on December 12, 2015, and entered into force on November 4, 2016. It aims to hold warming well below 2°C and to pursue efforts to limit it to 1.5°C above pre-industrial levels.

- **Source:** The Paris Agreement (UNFCCC)
- **In the posts:** No War! No Warming! (2015-12-13); Emergence of a New Global Civilization (2016-12-27); Wet'suwet'en and business as usual? NOT (2020-02-03); Wet'suwet'en 11.27.2021 (2021-11-27)

4. 2016–2020

Banking on Climate Change 2019 found that JPMorgan Chase provided $196 billion in fossil fuel financing from the Paris Agreement through 2018, and Banking on Climate Chaos 2021 found it provided nearly $317 billion from 2016 through 2020.

> ***Note:*** *Posts say 'over $195 billion'; the 2019 report figure is $196 billion.*

- **Source:** Global Banks, Led by JPMorgan Chase, Invested $1.9 Trillion in Fossil Fuels Since Paris Climate Pact (DeSmog); New Report: World's 60 Largest Banks Have Poured $3.8 Trillion Into Fossil Fuels (Sierra Club)
- **In the posts:** Wet'suwet'en and business as usual? NOT (2020-02-03); Defund the Coastal GasLink pipeline (2020-03-06); Frontline Indigenous Youth Washington, DC April 1 (2021-04-01)

5. 2017-04

The Sunrise Movement, a youth-led climate organization, was founded in April 2017.

- **Source:** Sunrise Movement (Wikipedia)
- **In the posts:** Sen Feinstein Rebuffs Youth and GND (2019-02-23); Indigenous led Green New Deal (2021-09-01); Buffalo Rebellion and Red/Green New Deal (2022-05-02)

6. 2017-06-01

President Trump announced on June 1, 2017 that the United States would withdraw from the Paris Agreement. The U.S. formally notified the UN on November 4, 2019, and the withdrawal took effect on November 4, 2020.

- **Source:** June 1: US Announced Withdrawal from Paris Climate Agreement (2017) (Today in Conservation)
- **In the posts:** Profits and Politics Trump the Earth (2017-06-02); We Are Not Terrorists (2018-08-30); Declare war on the United States? (2018-10-22)

7. 2017-08

Monsoon flooding in South Asia in the summer of 2017 killed more than 1,200 people across India, Bangladesh and Nepal and affected about 41 million people, at the same time as Hurricane Harvey's flooding in Texas.

- **Source:** South Asia floods: Affected areas and people impacted (Al Jazeera)
- **In the posts:** Harvey and the (near) Future (2017-08-31)

8. 2018

In 2018 wind generated about 6.5 percent of U.S. electricity and solar about 2.3 percent, together roughly 8 to 9 percent, according to EIA data.

> ***Note:*** *Posts say about 8 percent in early 2019; the figures are for calendar 2018, and the solar share includes small-scale solar.*

- **Source:** Renewables Provided 18% of U.S. Power Generation in 2018 (POWER Magazine)
- **In the posts:** Sunrise Movement succeeds in forcing discussion of climate policy (2019-03-23)

9. 2018-04

In April 2018 the monthly average carbon dioxide concentration at Mauna Loa Observatory reached 410.31 ppm, the first month in the record to exceed 410 ppm.

- **Source:** April monthly average exceeds 410 parts per million for the first time in recorded history (Scripps Institution of Oceanography, The Keeling Curve)
- **In the posts:** Coal Persists Despite Record CO2 levels (2018-12-06)

10. 2018-07-23

On July 23, 2018 the city of Kumagaya, northwest of Tokyo, reached 41.1°C (106°F), the highest temperature ever recorded in Japan.

- **Source:** Japanese heat wave pushes temperature to record (The Week (AP))
- **In the posts:** Climate Change on NBC (2018-08-04)

11. 2018-08-01

The State of the Climate in 2017 report (published August 1, 2018 in the Bulletin of the American Meteorological Society) found 2017 was the warmest non-El Niño year on record and the second or third warmest year overall; atmospheric CO2 averaged a record 405.0 ppm; and global sea level was rising about 1.2 inches per decade.

- **Source:** International report confirms 2017 was among the three warmest years on record (NOAA NCEI)
- **In the posts:** Climate Change on NBC (2018-08-04)

12. 2018-10-08

The IPCC Special Report on Global Warming of 1.5°C was approved in Incheon, South Korea and released on October 8, 2018. It found that limiting warming to 1.5°C requires global net CO2 emissions to fall about 45 percent from 2010 levels by 2030 and reach net zero around 2050.

> ***Note:*** *Posts paraphrase the finding as 'just 11 (or 12) years' to act, a popular reading of the 2030 target.*

- **Source:** Summary for Policymakers of IPCC Special Report on Global Warming of 1.5°C approved by governments (ScienceDaily (IPCC release))
- **In the posts:** Time to Panic (2019-02-19); Youth Climate Strike (2019-03-08); Global Climate Strikes (2019-03-15); US Army Predicts Collapse Within 20 Years (2019-10-28); Wet'suwet'en Updates and Solidarity Statements 2.14.2020 (2020-02-14)

13. 2018-11-13

On November 13, 2018 more than 200 Sunrise Movement activists occupied House Democratic leader Nancy Pelosi's office to demand climate action and a Green New Deal; 51 were arrested, and Representative-elect Alexandria Ocasio-Cortez visited the protesters.

- **Source:** 51 arrested for protesting Pelosi. Here's what they want (E&E News); Backed by Ocasio-Cortez, youth climate activists arrested at Pelosi's office (Nation of Change); Sunrise Movement (Wikipedia)
- **In the posts:** H.Res.109/S.Res.59 – Green New Deal (2019-02-11)

14. 2018-11-23

The Fourth National Climate Assessment, Volume II, was released on November 23, 2018 by the U.S. Global Change Research Program, with contributions from 13 federal agencies.

- **Source:** Fourth National Climate Assessment Volume II released (ScienceDaily (NOAA))
- **In the posts:** Coal Persists Despite Record CO2 levels (2018-12-06); As predicted, climate chaos continues (2019-05-23)

15. 2018-12-04

On December 4, 2018 the Iowa Utilities Board approved MidAmerican Energy's Wind XII wind project.

- **Source:** IUB Approves MidAmerican Wind XII Project, Sierra Club Responds
- **In the posts:** Coal Persists Despite Record CO2 levels (2018-12-06)

16. 2018-12-10 (corrected)

On December 10, 2018 more than 1,000 Sunrise Movement protesters held sit-ins in congressional offices on Capitol Hill, including Nancy Pelosi's, to demand a Green New Deal select committee; about 143 were arrested.

> ***Correction:*** *Post says about 138 were arrested; Roll Call reported about 143.*

- **Source:** Nearly 150 activists arrested in Green New Deal protest (Roll Call); Sunrise Movement (Wikipedia)
- **In the posts:** Green New Deal (2018-12-10); H.Res.109/S.Res.59 – Green New Deal (2019-02-11)

17. 2019 (corrected)

Studies published in 2019, including Brown University's Costs of War project, found the U.S. military to be the world's largest institutional consumer of hydrocarbons, emitting more than whole countries such as Portugal or Denmark.

> ***Correction:*** *Post calls the U.S. military 'the largest polluter in the world'; the cited reporting describes it as the largest institutional consumer of oil and emitter, not the largest polluter overall.*

- **Source:** Elephant in the room: The US military’s devastating carbon footprint (Al Jazeera)
- **In the posts:** The Fires in Gaza are the Fires in LA (2025-01-14)

18. 2019-02-07

The Green New Deal resolution was introduced on February 7, 2019: H.Res.109 by Representative Alexandria Ocasio-Cortez in the House and S.Res.59 by Senator Ed Markey in the Senate, where it was referred to the Environment and Public Works Committee. It was a non-binding resolution, not a bill.

- **Source:** H. Res. 109 (Introduced in House) (GovInfo); US SR59 | 2019-2020 | 116th Congress (LegiScan)
- **In the posts:** Resolution for Green New Deal Introduced (2019-02-07); Congressional Press Conference: Green New Deal (2019-02-09); H.Res.109/S.Res.59 – Green New Deal (2019-02-11); Green New Deal vote is back on Senate schedule (2019-03-06)

19. 2019-03

A powerful mid-March 2019 'bomb cyclone' caused record flooding on the Missouri River and its tributaries: 17 levees were breached or overtopped in Iowa, Nebraska and Missouri, Nebraska's Spencer Dam was destroyed, the Pine Ridge Reservation in South Dakota was cut off, and Nebraska alone estimated over $1.3 billion in preliminary damage.

- **Source:** River flooding inundates the Northern Plains in spring 2019 (NOAA Climate.gov)
- **In the posts:** Historic Flooding in Nebraska (2019-03-18); Midwest Climate Disaster (2019-03-26); Green New Deal addresses peace and justice (2019-03-29); Oglala Lakota Nation Flood Emergency Continues (2019-04-10)

20. 2019-03-14

Swedish climate activist Greta Thunberg was nominated for the Nobel Peace Prize in March 2019 by three Norwegian lawmakers, on the eve of the March 15, 2019 global school climate strike in more than 100 countries.

- **Source:** Greta Thunberg nominated for Nobel Peace Prize (Electrek)
- **In the posts:** Global Climate Strikes (2019-03-15); Youth Climate Strike (2019-03-03)

21. 2019-03-19

On March 19, 2019 Klawock, in southeast Alaska, reached 70°F, the earliest date any place in Alaska had reached that temperature.

- **Source:** Alaska hit 70 degrees the earliest ever (Coastal Care (CNN))
- **In the posts:** Midwest Climate Disaster (2019-03-26)

22. 2019-03-26

On March 26, 2019 the Green New Deal resolution brought to the Senate floor by Majority Leader Mitch McConnell failed 0–57, with 43 Democrats and independents voting 'present'.

- **Source:** AOC's Green New Deal dies in McConnell-led Senate vote (CNBC)
- **In the posts:** Senate Vote on Green New Deal (2019-02-13); Party Strategies and GND (2019-02-15); Green New Deal vote is back on Senate schedule (2019-03-06)

23. 2019-04-30

On April 30, 2019 a temporary flood barrier failed along the Mississippi River in downtown Davenport, Iowa, flooding about 15 blocks; the river crested near 22 feet, among the highest levels recorded there.

- **Source:** Davenport, Iowa flooded after barrier fails (CBS News)
- **In the posts:** Divest from Banks Funding Fossil Fuel (2019-05-01)

24. 2019-05-02

On May 2, 2019 the U.S. House passed the Climate Action Now Act (H.R.9) 231–190, with all Democrats and three Republicans voting yes. The bill barred funds for withdrawing from the Paris Agreement and required a plan to meet U.S. commitments under it.

- **Source:** Roll Call 184, H.R. 9 (Clerk of the U.S. House of Representatives); House passes bill opposing Trump Paris Agreement withdrawal (Climate Home News)
- **In the posts:** As predicted, climate chaos continues (2019-05-23)

25. 2019-09-20

The Global Climate Strike of September 20, 2019, timed three days before the UN Climate Action Summit, drew more than 4 million people worldwide; strikes during September 20–27 took place at about 4,500 locations in 150 countries. In Iowa, the strike gathered at the State Capitol in Des Moines.

- **Source:** September 2019 climate strikes (Wikipedia)
- **In the posts:** You're Invited: Global Climate Strike (2019-09-03); Iowa and Global Climate Strike (2019-09-17); Strike DC Press Release (2019-09-23)

26. 2020-08-10

On August 10, 2020 a derecho crossed eight Midwestern states over about 770 miles in 14 hours, with gusts estimated near 140 mph around Cedar Rapids, Iowa. It caused about $11 billion in damage, the costliest thunderstorm event in U.S. history, and damaged crops on roughly 14 million acres in Iowa.

- **Source:** A year ago, the Midwest was hit with the costliest thunderstorm event in U.S. history (NPR via KUNC); 2020 Midwest derecho (Wikipedia)
- **In the posts:** Exhilarated again (2020-08-11); Forget everything (2020-08-18); What kind of ancestor do you want to be? (2020-09-04); Racism as a Public Health Threat (2020-11-20)

27. 2020-12

The Copernicus Climate Change Service estimated that global warming reached 1.18°C above pre-industrial levels in December 2020, and that at the rate of the previous 30 years the 1.5°C level would be reached around 2034.

- **Source:** How close are we to reaching a global warming of 1.5°C? (Copernicus Climate Change Service)

- **In the posts:** Climate tipping points that might already have been reached (2021-04-29)

28. 2021-03

According to Rainforest Action Network's Banking on Climate Chaos 2021 report, JPMorgan Chase was the largest bank funder of fossil fuels since the Paris Agreement and provided $51.3 billion in fossil fuel financing in 2020.

- **Source:** The data is in: Chase, Citi and MUFG are hurling us all towards climate catastrophe (Rainforest Action Network); How much the largest banks have invested in fossil fuel (CNBC)
- **In the posts:** Frontline Indigenous Youth Washington, DC April 1 (2021-04-01)

29. 2021-08-09

The IPCC Working Group I report for the Sixth Assessment, released August 9, 2021, found that human greenhouse gas emissions have caused about 1.1°C of warming since 1850–1900 and that global temperature is expected to reach or exceed 1.5°C averaged over the next 20 years.

- **Source:** Climate change widespread, rapid, and intensifying – IPCC
- **In the posts:** Treaties Not Tar Sands (2021-08-25); PEOPLE VS. FOSSIL FUELS (2021-10-15); COP 26 and Beyond Coal and Gas Alliance (2021-11-14)

30. 2021-10-31–2021-11-12

The UN climate conference COP26 was held in Glasgow, Scotland, from October 31 to November 12, 2021 (UK presidency in partnership with Italy) and ended with the Glasgow Climate Pact.

> ***Note:*** *Negotiations ran a day past the scheduled close; the pact was agreed on November 13, 2021.*

- **Source:** Glasgow Climate Change Conference - October-November 2021 (UNFCCC)
- **In the posts:** Capitalism must be reprogrammed with mutual aid (2021-11-02); COP 26 and continued colonial capitalism (2021-11-05); COP26 Not Nearly Enough (2021-11-12); COP 26 and Beyond Coal and Gas Alliance (2021-11-14)

31. 2022-03-31

On March 31, 2022 President Biden announced the release of one million barrels of crude oil a day from the Strategic Petroleum Reserve for six months (about 180 million barrels), citing supply disruption from Russia's war on Ukraine.

- **Source:** DOE Announces Additional Notice of Sale of Crude Oil from the Strategic Petroleum Reserve (U.S. Department of Energy)
- **In the posts:** Mutually exclusive (2022-04-01); Banking on climate chaos (2022-04-06); Dangerous disconnect (2022-06-15)

32. 2022-04-04

On April 4, 2022 ExxonMobil announced a $10 billion final investment decision for its Yellowtail oil development offshore Guyana, its fourth and largest project in the Stabroek Block, designed to produce about 250,000 barrels a day.

- **Source:** ExxonMobil makes final investment decision on fourth Guyana offshore project
- **In the posts:** Investing in new fossil fuel infrastructure is moral and economic madness (2022-04-08)

33. 2022-04-04

The IPCC released its Sixth Assessment Working Group III report on mitigating climate change on April 4, 2022.

- **Source:** IPCC Sixth Assessment Report (Wikipedia)
- **In the posts:** Banking on climate chaos (2022-04-06); Investing in new fossil fuel infrastructure is moral and economic madness (2022-04-08); Earth Day Rally (2022-04-21)

34. 2022-04-04

When the IPCC Working Group III report was released on April 4, 2022, UN Secretary-General António Guterres said that 'investing in new fossil fuels infrastructure is moral and economic madness.'

> ***Note:*** *UN and IPCC pages could not be loaded; the quotation was confirmed in a published commentary reproducing his remarks.*

- **Source:** The World's Top Diplomat Has Had It Up to Here (The Crucial Years (Bill McKibben))
- **In the posts:** Investing in new fossil fuel infrastructure is moral and economic madness (2022-04-08); Another nail in our coffin (2022-04-18)

35. 2022-08

The Inflation Reduction Act required the Interior Department to hold specified offshore oil and gas lease sales, including in the Gulf of Mexico, as a condition linked to new wind and solar leasing on federal lands, a provision that came from the compromise with Senator Joe Manchin.

- **Source:** Inflation Reduction Act (Wikipedia); The climate effects of the Inflation Reduction Act (The Week)
- **In the posts:** What should be non-negotiable (2023-05-30)

36. 2022-08-16

President Biden signed the Inflation Reduction Act on August 16, 2022. Its climate and energy provisions included raising the 45Q carbon-capture tax credit (for example from $50 to $85 per ton for stored industrial CO2 and to $180 for direct air capture) and lowering eligibility thresholds.

- **Source:** Inflation Reduction Act (Wikipedia); The Inflation Reduction Act Includes Significant Benefits for the Carbon Capture Industry (Gibson Dunn)
- **In the posts:** Buffalo Rebellion Community Call: Carbon Pipeline Resistance (2022-12-19); The Great Carbon Boondoggle (2023-01-18); Carbon Capture update 4/4/2023 (2023-04-04)

37. 2023-03-13

On March 13, 2023 the Biden administration approved ConocoPhillips' Willow oil project in Alaska with three drill pads; it is expected to produce up to 600 million barrels over 30 years.

- **Source:** Willow project (Wikipedia)
- **In the posts:** Willow Project (2023-03-16); What should be non-negotiable (2023-05-30)

38. 2024-05-28

During India's 2024 heat wave, Churu in Rajasthan recorded 50.5°C (122.9°F) on May 28, 2024, the highest temperature in India in eight years.

- **Source:** 2024 Indian heat wave (Wikipedia)
- **In the posts:** The brink of ‘survivability’ (2024-06-01)

39. 2024-07

July 2024 was California's hottest month on record, with a statewide average temperature of 81.7°F according to NOAA data.

- **Source:** July was California's hottest month ever, as climate warms to dangerous new extremes (Los Angeles Times (via AOL))
- **In the posts:** Climate Graphs 2 (2024-08-19)

40. 2025-01-07

The Palisades and Eaton wildfires in Los Angeles began on January 7, 2025, driven by strong Santa Ana winds. They destroyed or damaged more than 18,000 structures and officially killed 31 people.

- **Source:** January 2025 Southern California wildfires (Wikipedia)
- **In the posts:** The Fires in Gaza are the Fires in LA (2025-01-14); Desolate landscapes (2025-01-19)
